Supplementary Material for "Reliable cross-view neighborhood relation transfer and high-order semantic structure mining for incomplete multi-view partial multi-label classification"

S1 Additional Diagnostic Analysis of Reliable Cross-View Completion

S1.1 Diagnostic protocol

We conduct additional post-hoc diagnostic analyses on Corel5k and Pascal07 under the random dual-incomplete setting with 50% missing views and 50% missing labels. All statistics are pooled over ten folds using missing-sample–target-view completion cases as the statistical units. For each missing target view, we record the availability of transferred candidates, their accumulated source-view distances, and the normalized reconstruction weights. Weight-related statistics are calculated only for cases with at least one valid candidate.

S1.2 Diagnostic metrics

Let $q_{i,v}$ denote the number of valid transferred candidates for sample i missing in target view v. We report the average candidate number, the percentage of cases with $q_{i,v} = 0$, and the percentage of cases satisfying $0 < q_{i,v} < k$.

To examine whether very small accumulated distances cause excessive weight concentration, we report the minimum and 5th percentile of all valid-candidate accumulated distances, denoted by $D_{\min}$ and $D_{5\%}$, respectively. For each completion case with at least one valid candidate, we calculate the maximum normalized reconstruction weight as

$$w_{\max}^{(i,v)} = \max_{j \in \mathcal{N}_{i,v}} \omega_{i,j}, \tag{S1}$$

and report its mean and 95th percentile.

The effective number of contributing neighbors is defined as

$$N_{\text{eff}}^{(i,v)} = \frac{1}{\sum_{j \in \mathcal{N}_{i,v}} \omega_{i,j}^2}. \tag{S2}$$

A value close to k indicates that multiple neighbors jointly contribute to reconstruction, whereas a value close to one indicates that the reconstruction is dominated by a single neighbor. We additionally report N_{eff}/k to facilitate comparison between datasets using different values of k.

S1.3 Results and discussion

Table S1: Diagnostic analysis of reliable cross-view completion under the random dual-incomplete setting with 50% missing views and 50% missing labels.

Dataset	k	Avg. Cand.	No Cand. (%)	$0 < q < k$ (%)	$D_{\min}/D_{5\%}$	$\overline{w}_{\max}/w_{\max}^{95\%}$	N_{eff}	N_{eff}/k (%)
Corel5k	5	5.722	0.685	35.572	1.527/8.567	0.280/0.504	4.253	85.052
Pascal07	9	9.878	0.054	39.261	0.024/8.721	0.153/0.333	7.684	85.373

As shown in Table S1, the proportions of completion cases without any valid transferred candidate are only 0.685% on Corel5k and 0.054% on Pascal07. Thus, at least one valid candidate is available for 99.315% and 99.946% of the evaluated cases, respectively. Meanwhile, 35.572% and 39.261% of the cases contain at least one but fewer than k valid candidates, confirming the practical relevance of adaptively using all available candidates.

No exactly zero accumulated distance is observed in the ten-fold diagnostic results. The minimum accumulated distances are 1.527 and 0.024 on Corel5k and Pascal07, whereas their 5th percentiles are 8.567 and 8.721, respectively. This indicates that very small distances occur only in the extreme lower tail of the candidate-distance distribution.

More importantly, the mean maximum normalized weights are 0.280 and 0.153, while their 95th percentiles are 0.504 and 0.333. The corresponding effective numbers of contributing neighbors are 4.253 for $k = 5$ and 7.684 for $k = 9$, accounting for 85.052% and 85.373% of the nominal neighborhood sizes. These results indicate that very small accumulated distances do not generally cause the reconstruction weights to collapse onto a single candidate. Nevertheless, extremely sparse cross-view overlap may still reduce the available transferable neighborhood information.

S2 Robustness under four non-random missing mechanisms

To evaluate robustness beyond random missingness, we test four non-random mechanisms on Corel5k and Pascal07 using the same first three folds for all methods. Table S2 reports the mean and standard deviation.

In the view-specific setting, the six views have missing rates of 20%, 40%, 50%, 50%, 60%, and 80%, respectively, resulting in an average missing-view rate of 50%. In the block-wise setting, the samples are divided into four blocks, and each block misses a fixed subset of three views, also yielding an overall missing-view rate of 50%. Both settings use a 50% missing-label rate. The label-dependent setting retains random 50% view missingness and divides the labels into head, middle, and tail groups according

Table S2: Comparison results under four non-random missing mechanisms on Corel5k and Pascal07. The results are reported as the mean and standard deviation over the same first three folds.

SETTING	METRIC	DICNet	DSPNF	RANK	OURS
View-specific	AP↑	$0.341_{0.004}$	$\underline{0.402}_{0.002}$	$0.391_{0.004}$	$\mathbf{0.451}_{0.004}$
	1-HL↑	$0.987_{0.000}$	$\underline{0.988}_{0.000}$	$0.987_{0.000}$	$\mathbf{0.988}_{0.000}$
	1-RL↑	$0.876_{0.003}$	$0.892_{0.003}$	$\underline{0.895}_{0.001}$	$\mathbf{0.917}_{0.002}$
	AUC↑	$0.879_{0.003}$	$0.895_{0.003}$	$\underline{0.898}_{0.002}$	$\mathbf{0.921}_{0.001}$
Block-wise	AP↑	$0.337_{0.004}$	$\underline{0.363}_{0.005}$	$0.229_{0.006}$	$\mathbf{0.428}_{0.011}$
	1-HL↑	$0.987_{0.000}$	$\underline{0.987}_{0.000}$	$0.987_{0.000}$	$\mathbf{0.988}_{0.000}$
	1-RL↑	$0.867_{0.002}$	$\underline{0.883}_{0.001}$	$0.826_{0.004}$	$\mathbf{0.914}_{0.002}$
	AUC↑	$0.871_{0.001}$	$\underline{0.886}_{0.001}$	$0.830_{0.004}$	$\mathbf{0.916}_{0.001}$
Label-dependent	AP↑	$0.360_{0.006}$	$0.410_{0.007}$	$\underline{0.422}_{0.010}$	$\mathbf{0.459}_{0.005}$
	1-HL↑	$0.987_{0.000}$	$0.988_{0.000}$	$\underline{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$
	1-RL↑	$0.872_{0.002}$	$0.893_{0.003}$	$\underline{0.904}_{0.003}$	$\mathbf{0.916}_{0.001}$
	AUC↑	$0.876_{0.003}$	$0.897_{0.002}$	$\underline{0.907}_{0.003}$	$\mathbf{0.919}_{0.001}$
One-view	AP↑	$0.225_{0.002}$	$\underline{0.305}_{0.008}$	$0.245_{0.006}$	$\mathbf{0.335}_{0.004}$
	1-HL↑	$0.987_{0.000}$	$0.987_{0.000}$	$\underline{0.987}_{0.000}$	$\mathbf{0.987}_{0.000}$
	1-RL↑	$0.804_{0.004}$	$\underline{0.858}_{0.005}$	$0.838_{0.008}$	$\mathbf{0.876}_{0.005}$
	AUC↑	$0.807_{0.004}$	$\underline{0.861}_{0.005}$	$0.842_{0.009}$	$\mathbf{0.880}_{0.004}$

(a) Corel5k

SETTING	METRIC	DICNet	DSPNF	RANK	OURS
View-specific	AP↑	$0.544_{0.006}$	$0.456_{0.011}$	$\underline{0.578}_{0.007}$	$\mathbf{0.595}_{0.009}$
	1-HL↑	$0.933_{0.001}$	$0.927_{0.001}$	$\underline{0.934}_{0.001}$	$\mathbf{0.935}_{0.001}$
	1-RL↑	$0.819_{0.004}$	$0.754_{0.005}$	$\underline{0.841}_{0.003}$	$\mathbf{0.852}_{0.001}$
	AUC↑	$0.842_{0.002}$	$0.780_{0.004}$	$\underline{0.860}_{0.001}$	$\mathbf{0.872}_{0.003}$
Block-wise	AP↑	$\underline{0.549}_{0.007}$	$0.447_{0.015}$	$0.457_{0.007}$	$\mathbf{0.570}_{0.008}$
	1-HL↑	$\underline{0.932}_{0.001}$	$0.927_{0.001}$	$0.925_{0.001}$	$\mathbf{0.935}_{0.001}$
	1-RL↑	$\underline{0.831}_{0.002}$	$0.736_{0.019}$	$0.737_{0.010}$	$\mathbf{0.837}_{0.002}$
	AUC↑	$\underline{0.852}_{0.000}$	$0.767_{0.017}$	$0.772_{0.003}$	$\mathbf{0.858}_{0.002}$
Label-dependent	AP↑	$0.537_{0.013}$	$0.455_{0.015}$	$\underline{0.563}_{0.007}$	$\mathbf{0.581}_{0.011}$
	1-HL↑	$0.932_{0.000}$	$0.927_{0.001}$	$\underline{0.933}_{0.001}$	$\mathbf{0.934}_{0.001}$
	1-RL↑	$0.815_{0.005}$	$0.749_{0.007}$	$\underline{0.829}_{0.003}$	$\mathbf{0.845}_{0.006}$
	AUC↑	$0.839_{0.003}$	$0.777_{0.007}$	$\underline{0.851}_{0.003}$	$\mathbf{0.867}_{0.003}$
One-view	AP↑	$0.447_{0.002}$	$0.434_{0.010}$	$\underline{0.451}_{0.004}$	$\mathbf{0.496}_{0.005}$
	1-HL↑	$0.925_{0.001}$	$\underline{0.926}_{0.000}$	$0.924_{0.001}$	$\mathbf{0.930}_{0.000}$
	1-RL↑	$0.733_{0.004}$	$0.717_{0.003}$	$\underline{0.743}_{0.005}$	$\mathbf{0.789}_{0.006}$
	AUC↑	$0.761_{0.008}$	$0.746_{0.002}$	$\underline{0.775}_{0.003}$	$\mathbf{0.812}_{0.005}$

(b) Pascal07

to their positive frequencies in the training set, with missing-label rates of 30%, 50%, and 70%, respectively. In the one-view-observed setting, each sample retains only one view, corresponding to an 83.3% missing-view rate, while the missing-label rate remains 50%.

As shown in Table S2, ReCHSM achieves the best or tied-best performance under all four non-random missing mechanisms on both datasets. The relative difficulty of these protocols varies with the dataset characteristics. On Corel5k, the label-dependent setting yields relatively strong results for most methods. A possible reason is that Corel5k contains 260 labels and has a sparse label distribution; preserving more supervision for frequent labels can provide relatively stable semantic guidance. In contrast, Pascal07 contains only 20 object categories and has a more compact label space, so the view-specific setting can be more favorable when relatively informative views are preserved with lower missing rates. The one-view-observed setting is consistently the most difficult case for all methods because its missing-view rate reaches 83.3% and most cross-view complementary information is unavailable.

The one-view-observed setting also substantially restricts candidate-based completion because a transferred neighbor must satisfy the observation requirements of both the source and target views, greatly reducing the availability of valid candidates when each sample retains only one view. It therefore serves as an extreme stress test in which cross-view completion is largely weakened. ReCHSM nevertheless achieves the best or tied-best results, indicating that observed-view representation learning, high-order structure mining, and quality-aware fusion remain effective when cross-view complementarity is highly limited.